

Dr. Zoe Molitor (*she/they*)
Earth and Environmental Sciences
Boston College

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Education and Employment

- 2026-current Postdoctoral Researcher
 EES, Boston College, Chestnut Hill, MA
- 2024-2026 NSF EAR Postdoctoral Fellow
 EPS, Yale University, New Haven, CT
- 2018-24 Ph.D. Geology
 EAPS, MIT, Cambridge, MA
- 2015-18 B.S. Geology and Geophysics
 Purdue University, West Lafayette, IN

Publications

Molitor, Z., Link, F., Long, M., Ague, J. (2026). Extreme Moho Relief Preserved in Ohio's Crust. *GSA Today*, 36, 4-9.

Liu, Z., Jagoutz, O., Rezeau, H., Zhang, H. L. L., Klein, B., **Molitor, Z.**, ... & Zhu, D. C. (2026). A whole-rock geochemistry-based semi-quantitative proxy for magmatic oxygen fugacity. *Earth and Planetary Science Letters*, 673, 119742.

Molitor, Z., Ague, J., Graham, B. (2025). Bedrock Geology of NE Connecticut: A Transect from the Bronson Hill Anticlinorium to Putnam-Nashoba Terrane. In Rooney, A.D. and Long, M.D. (Eds.), *Guidebook for field trips in southwestern Connecticut* (Trip D, pp. 86-112). Geology Documents. https://digitalmaine.com/geo_docs/204/.

Molitor, Z., Jagoutz, O., Moser, A., Cruz-Uribe, A., & Cross, A. (2025). Permo-Carboniferous crustal structure and state of strain in the New England Appalachians, USA. *Geosphere*, 21(6), 991-1019.

Liu, Z., Jagoutz, O., **Molitor, Z.**, Ramezani, J., & Zhu, D. C. (2025). Crustal section in the New England Appalachians records polybaric fractional crystallization-driven magma differentiation. *Earth and Planetary Science Letters*, 671, 119647.

Molitor, Z., Mittal, T., & Jagoutz, O. (2024). The viscosity of a partially molten layer in a paleo-orogenic plateau. *Earth and Planetary Science Letters*, 648, 119060.

Brown Krein, S., **Molitor, Z.**, & Grove, T. L. (2021). ReversePetrogen: A Multiphase Dry Reverse Fractional Crystallization-Mantle Melting Thermobarometer Applied to 13,589 Mid-Ocean Ridge Basalt Glasses. *Journal of Geophysical Research: Solid Earth*, 126(8), e2020JB021292.

Molitor, Z., Jagoutz, O., Cross, A., Moser, A. (**In Prep**), Testing models of orogenic conjugate shear in the New England Appalachians.

Invited Talks

November 2025—Lahmont Doherty Earth Observatory Geodynamics Seminar—Extreme Moho Relief Beneath the State of Ohio

Conference Abstracts

Molitor, Z., Ague, J., Graham, B., Eckert, J., Cruz-Urbe, A. (2026). Spatio-Temporal Controls on Orogenic Thermal Structure and Evolution in the Southern New England Appalachians. Abstract 29-5 presented at 2026 Northeast GSA, 22-23 Mar.

Graham, B., **Molitor, Z.**, Ague, J. (2026). Assessing the Impact of High Temperature Metamorphism and Partial Melting on the Viscosity of the Tatnic Hill Formation. Abstract 284-7 presented at 2025 GSA Connects, 19-22 Oct.

Molitor, Z., Ague, J., Graham, B., Eckert, J. (2025). Spatio-Temporal Controls on Orogenic Thermal Structure and Evolution in the Southern New England Appalachians. Abstract 284-7 presented at 2025 GSA Connects, 19-22 Oct.

Molitor, Z., Jagoutz, O., Cross, A., Cruz-Urbe, A., Moser, A. (2024). Constraining the 3D Geometry of Conjugate Orogenic Strike-Slip Shear Zones in the New England Appalachians. Abstract 20-6 presented at 2024 Northeast Geological Society of America Meeting, 17-19 Mar.

Molitor, Z., Royden, L., Jagoutz, O (2022). Testing Models of Conjugate Faulting in the Late Paleozoic New England Appalachian Orogen, Abstract (T55B-0068) presented at 2022 AGU Fall Meeting, 12-16 Dec.

Molitor, Z., Royden, L., Jagoutz, O (2021). A New Model for Dynamic and Thermal Compensation around Mantle Plumes: Implications for Volume and Heat Flux, Abstract (DI25B-0027) presented at 2021 AGU Fall Meeting, 13-17 Dec.

Molitor, Z., Jagoutz, O., Royden, L., Brown, S., Port, G., Dogru, A., and Keller, J. (2020). Mantle Plume – Spreading Ridge Interaction: A Comparative Study of the Red Sea and Reykjanes Ridges, Abstract (EGU2020-11138) presented at 2020 EGU General Assembly, 4-8 May 2020.

Teaching Experience and Outreach

Fall 2026	Adjunct Professor—Structural Geology, EESC 3385
2021-22	Graduate Teaching Assistant, 12.002 MIT, Spring 2021 and 2022
2017	Undergraduate Teaching Assistant, EAPS 12000 Purdue University
2025	Co-Organizer of Earth Science Week-Long Workshop – Pathways Summer Scholars High School Program, Yale University
2024	Group Leader-Chemistry Rotation During West Campus Science Festival, Yale University
2022-24	Administrative Member of LINK12, K12 Outreach Program at EAPS MIT
2022	Letters to a PreScientist, Pen Pal program with K12 Students
2022	Field Trip on the Geology of the White Mountains for DEAPS: Freshman Orientation
2021	Boston826: Hydraulic Fracturing Demonstration and Interview with Local K12 Students
2020	Field Trip on the Geology of Yellowstone for DEAPS: Freshman Orientation

Service

2024-2026	Founder and Lead Organizer of LGBTQ+ Working Group, Yale University
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2024-2026 WEPSY (Women and nonbinary people in EPS) Organizing Committee, Yale University
2024-2026 Member of EPS Department IDEA Committee, Yale University

Mentoring

Summer 2025 Brynn Graham (Undergraduate - Mt. Holyoke College), SURF Program at Yale University
And Spring 2026

Field Experience

Spring 2026 Nahant and Salem, MA: *3 Days*
Fall 2025 Northeast Intercollegiate Geologic Conference (NEIGC, Connecticut), Field Trip Leader
Fall 2024 Connecticut, Massachusetts, and New Hampshire: *6 Weeks*
June 2023 Sicily, Italy (MIT Department Field Trip): *1 Week*
Summer 2022 New England Appalachians (NH, MA, and CT): *4 Weeks*
Jan. 2022 Mojave Desert (MIT Field Camp): *2 Weeks*
Dec. 2021 Vermont and New Hampshire: *1 Week*
Nov. 2021 Newfoundland, Canada: *2 Weeks*
Jun. 2021 New Hampshire: *1 Week*
Fall 2020 New Hampshire: *4 Weeks*
Jan. 2019 Mojave Desert (MIT Field Camp): *2 Weeks*
Jun. 2018 Sardinia, Italy (Undergraduate Field Camp): *4 Weeks*
Mar. 2018 Utah (Undergraduate Field Methods): *1 Week*

Technical Skills

Programming Languages

- Python, MATLAB (MTEX)

Computer Programs

- ArcGIS, Adobe Illustrator and Inkscape, Perple_X, TheriakDomino, Microsoft Suite

Laboratory Analysis

- SEM/EPMA (Electron Probe Microanalyzer), SEM/EBSD (Electron Backscatter Diffraction), Heavy Mineral Separation, LA-ICP-MS